TILOSCHAN KARKI

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Education

Texas State University - Honors College **San Marcos, TX**
B.S. in Computer Science; Minors in Data Analytics and Applied Mathematics *May 2026*
GPA: 3.74/4.00
Coursework: Machine Learning, Data Mining & Information Retrieval, Computer Architecture, Computer Networks, Systems Security

Research Interests

Machine Learning Systems; GPU/HPC Resource Management; Cluster Scheduling; Workload Characterization; Distributed Systems; ML-Guided Systems Optimization; HPC and Parallel Computing

Research Experience

Intent-Aware Resource Prediction for GPU Cluster Scheduling *Independent Research - Work in Progress* **2026-Present**

- Formulated a research question examining whether structured user intent adds predictive value beyond job metadata and workload history for pre-execution GPU, memory, and runtime estimation.
- Synthesized SchedMate, Liquid, Pollux, Lucid, and related HPC resource-prediction work in a comparison matrix; refined the proposed contribution around measuring the marginal value of intent rather than broadly claiming semantic-aware scheduling as novel.
- Developed an evaluation plan for metadata-, history-, and intent-based ablations using public workload traces and trace-driven simulation; selected prediction error, GPU utilization, wasted GPU time, queue time, job failures, and scheduling overhead as planned metrics.

IoT Malware Detection with Scenario-Based Evaluation *Python, scikit-learn, CTU-IoT-23* **2025**

- Processed 10+ GB of raw Zeek logs and built a reproducible pipeline that preserved capture-level separation, preventing leakage from random record-level splits.
- Evaluated Decision Tree and Random Forest baselines on held-out devices, attacks, and capture environments using accuracy, precision, recall, F1, ROC curves, and confusion matrices.
- Found that the Decision Tree achieved approximately 90% accuracy on held-out scenarios, while the Random Forest achieved approximately 42%, suggesting substantially weaker cross-scenario generalization by the ensemble model.

Teaching Experience

Physics Undergraduate Instructional Assistant **Texas State University**
Undergraduate instruction and quantitative problem solving *Aug. 2022-May 2026*

- Supported undergraduate students across instructional and problem-solving activities by diagnosing conceptual errors and explaining quantitative methods step by step.
- Adapted technical explanations to students with different levels of preparation, emphasizing the physical reasoning behind mathematical procedures rather than supplying answers alone.

Systems & Machine Learning Projects

ScholarGraph *Neo4j, Node.js, React, Solidity, Polygon* **2025**

- Co-designed a scholarly knowledge graph linking papers, authors, institutions, topics, datasets, and citations; ingested metadata for 200+ papers through the Crossref API.
- Built Neo4j/Express query services and a Solidity verification layer that recorded DOI hashes on Polygon Amoy for consistency checks against blockchain records.
- Demonstrated end-to-end verification across a React interface, graph database, REST API, and deployed smart contract in a two-developer team.

Tfolio *Django REST, React, PostgreSQL, AWS* **2025-Present**

- Built a structured research and project portfolio with Django models, REST APIs, and a React frontend, replacing hard-coded content with an extensible content-management workflow.
- Deployed across Vercel and AWS EC2/RDS using Gunicorn and Nginx; implemented portable configuration and data-recovery workflows across hosted PostgreSQL, local SQLite, and Django fixtures.

Amazon Review Sentiment Analysis *Python, TF-IDF, SVM, Logistic Regression, NumPy* **2025**

- Compared SVM, Logistic Regression, and Naive Bayes on approximately 105,000 appliance reviews using a shared 5,000-feature TF-IDF pipeline and class-specific evaluation.
- Achieved 94.0% overall accuracy with SVM while identifying weaker negative-class F1 performance (75%), distinguishing aggregate accuracy from minority-class behavior.

Technical Skills

Languages: Python, JavaScript, SQL, C++, Java, Solidity
ML/Data: scikit-learn, pandas, NumPy, Jupyter, TF-IDF, classification and error analysis
Research Methods: Literature synthesis, ablation design, scenario-based evaluation, trace-driven simulation planning, LaTeX
Systems/Cloud: Linux, AWS EC2/RDS, Docker, Gunicorn, Nginx, Vercel
Databases/Search: PostgreSQL, SQLite, Neo4j, Elasticsearch, Crossref API
Frameworks/Tools: Django REST Framework, Flask, React, Node.js, Streamlit, Git, GitHub

Achievements & Activities

Recognition: Texas State Presidential Scholarship; Dean's List (2022-2026); Certificate of Excellence, St. Xavier's School (2021)
Activities: Member, Institute of Electrical and Electronics Engineers (IEEE)